AVOIDING ANALYSIS TRAP

CHALLENGES

DATA CHALLENGES

ANALYSIS CHALLENGES

LIMITATIONS OF DATA

- Missing Data: Incomplete records can lead to biased or incorrect models.
- Small Sample Sizes: Insufficient data might fail to capture the variability of the entire population.
- Data Quality Issues: Errors, duplicates, or inconsistencies in the data can distort results.

IMPUTATION TECHNIQUES

CHOOSING APPRORIATE SAMPLING TECHNIQUES

DATA CLEANING

DEALING WITH OUTLIERS

OUTLIERS CAUSED BY OUTSIDE FORCES

OULIERS CAUSED BY ERRORS IN THE SYSTEM

Solution:

- Detect Outliers: Use methods like boxplots, z-scores, or interquartile range (IQR).
- Treat Outliers: Remove, transform, or assign appropriate values based on the context and business understanding.
- Two analysis approaches: It's recommended to conduct two analyses

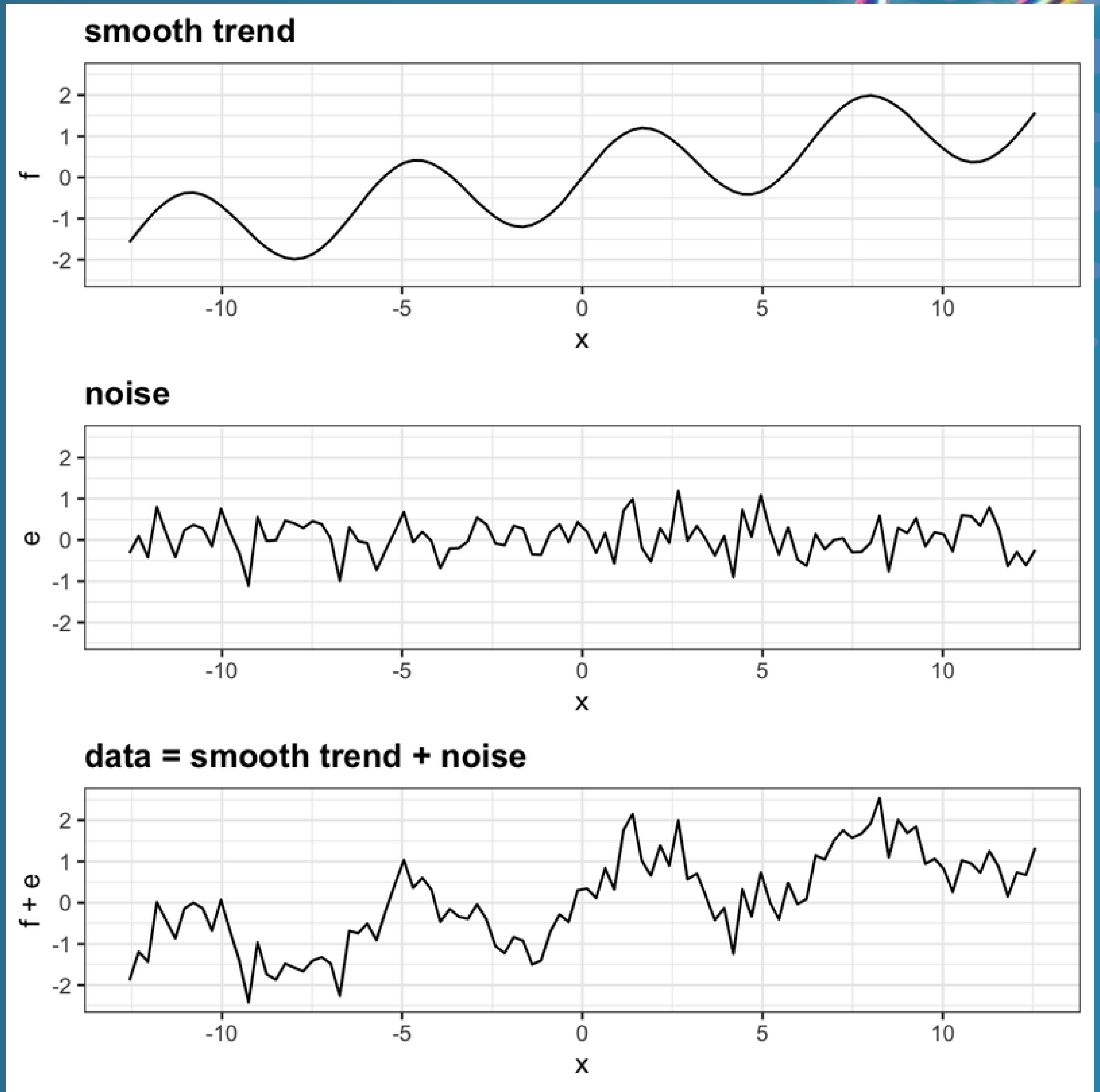
Comparing the results helps understand their impact on the analysis.

- Domain expertise is crucial: Business domain experts can provide valuable insights

DATA SMOOTHING

Data smoothing refers to a statistical approach of eliminating outliers from datasets to make the patterns more noticeable.

It consist of two parts
1. Core data points (trends)
2. Deviations (noise)



ASSUMPTIONS TO REDUCE NOISE

1. Fluctuations in a data to be noise
2. That noise should be in short duration
3. Data fluctuations won't affect the underlying trends

METHODS

1. Random Walk

2. Moving Average (MA)

3. Exponential Smoothing
a) Simple Exponential Smoothing (SES)
b) Linear
c) Seasonality

DATA SMOOTHING

ADVANTAGES

1. Easy to implement
2. Helps to identify trends
3. Helps to expose patterns in the data
4. Eliminate data points that you've decided aren't of interest
5. Generate nice smooth graphs

DISADVANTAGES

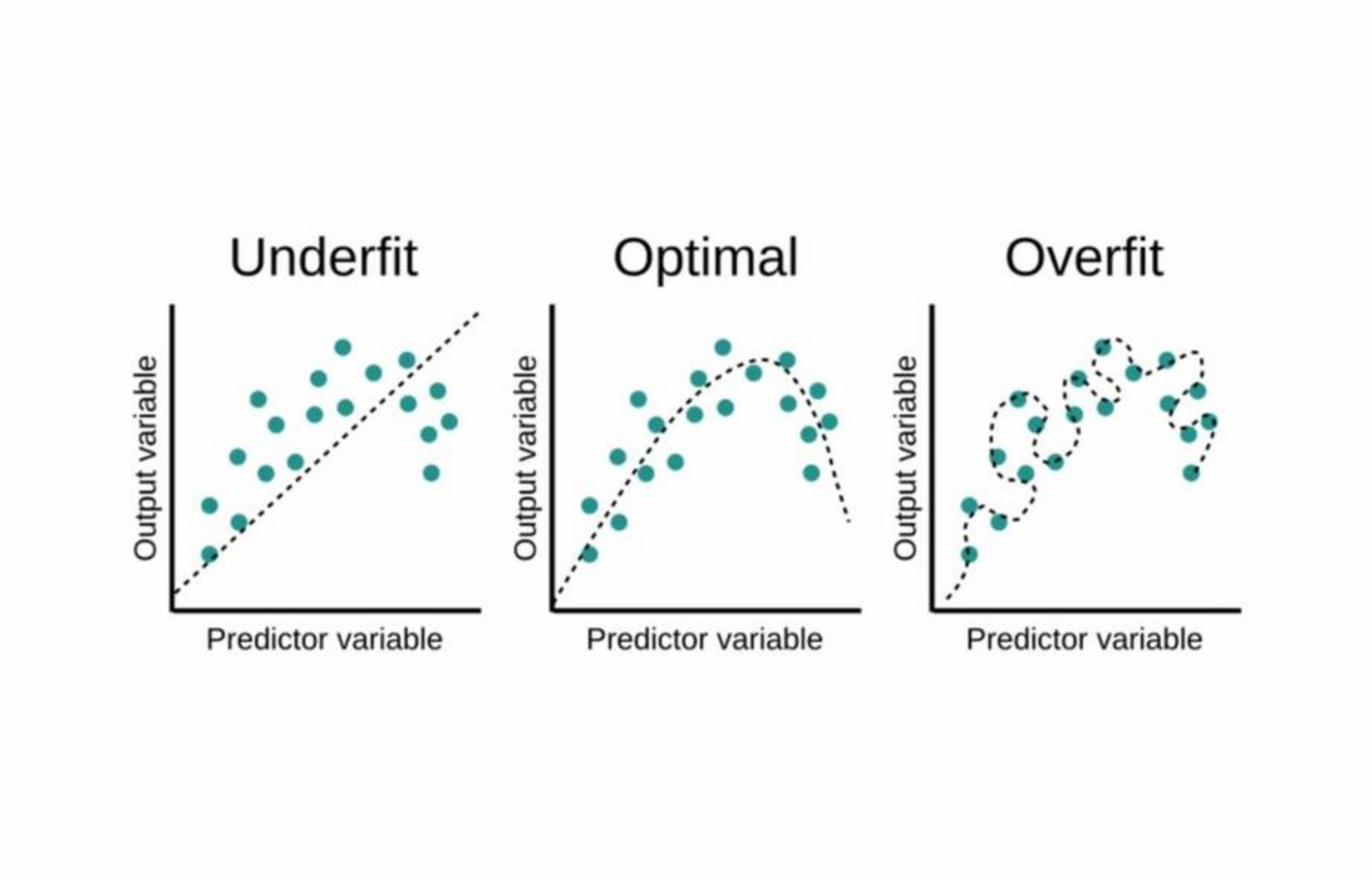
1. Eliminate valid data points from extreme events.
2. Led to inaccurate predictions
3. May be vulnerable to significant disruptions from outliers within the data
4. May result in a major deviation from the original data

CURVE FITTING

Definition

The process of building a mathematical function that best represents the relationship between variables in the dataset.

Examples:
Overfitting: A model that captures even minor noise in the data, leading to poor generalization.
Underfitting: A model that is too simple and fails to capture the complexity of the data.



Impact: Overfitting leads to models that perform well on training data but poorly on new data. Underfitting results in weak predictions overall.

Solution:

Regularization: Techniques like L1 or L2 regularization to prevent overfitting.
Cross-Validation: Ensuring the model works well on both training and unseen data.

MINIMIZING ASSUMPTIONS

1. AVOID UNNECESSARY ASSUMPTIONS
2. FOCUS ON BUSINESS KNOWLEDGE
3. TEST AND IMPROVE ASSUMPTIONS
4. APPLY OCCAM'S RAZOR*
5. BALANCE BETWEEN NECESSARY VARIABLES AND UNNECESSARY VARIABLES

* - When there are multiple possible explanations or models, choose the simplest one that still fits the data.

ANALYSIS CHALLENGES

1.SUPERVISED ANALYTICS

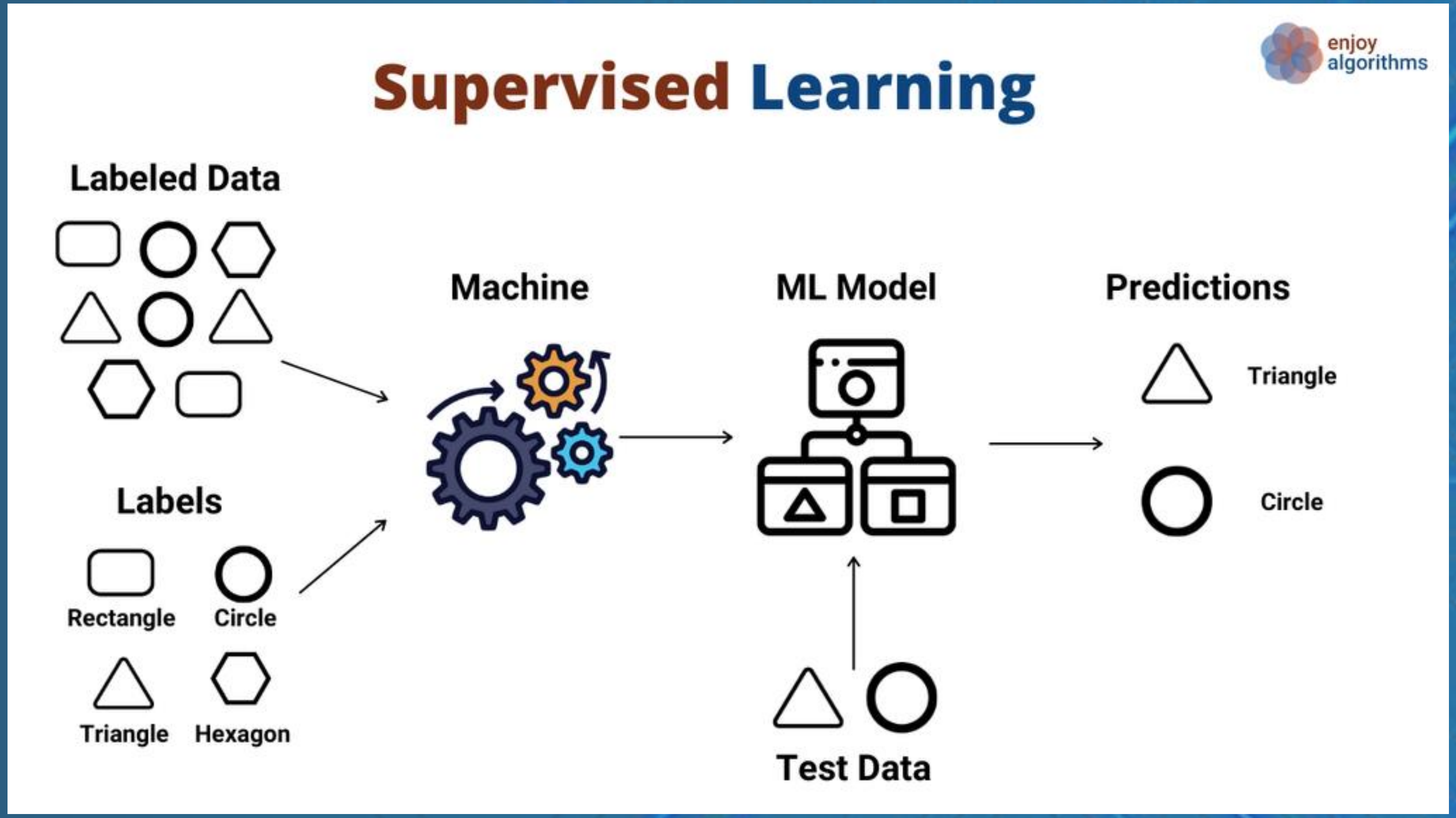
1. **Training:** The model is trained on a dataset with input and the corresponding output
2. **Learning:** The model learns to identify patterns and relationships between them
3. **Prediction:** The model can predict the output for unseen data based on the patterns.

Advantages

- Analyst Control
- Known Classifications
- Easy Error Resolution

Disadvantages

- Reinforced Mistakes
- Limited Population Representation
- Difficulty Detecting New Classes
- Overlapping Cluster



2. AVOIDING SINGLE ANALYSIS DEPENDENCE

Why is relying on one analysis problematic?

- "One-size-fits-all" is rarely applicable
- Limited Perspective
- Increased Risk of Error
- Missed Opportunities

Strategies for Avoiding Single Analysis Dependence

- Run Multiple Analyses
- Iterative Testing
- Compare Models
- Evaluate

DESCRIBING MODEL LIMITATIONS

1. DATA NONLINEARITY
2. VARIABLE CORRELATIONS
3. VARIABLE INDEPENDENCE
4. OVERFITTING DUE TO DATA SCOPE

AVOIDING NON-SCALABLE MODELS

- Non-Scalable models work with small data but struggle as data grows.
- **Key Issues:**
  - Overfitting: Models too complex for small datasets.
  - Feature Overload: Too many features slow down performance.
  - Manual Scalability: Reliance on manual tuning instead of automated processes.
- **How to Avoid:**
  - Scalable Models: Use models like Random Forest, XGBoost, or Neural Networks.
  - Regularization: Apply L1/L2 regularization to control complexity.
  - Distributed Computing: Leverage tools like Apache Spark.
  - Dimensionality Reduction: Use PCA or feature selection for efficient data handling.
  - Cross-Validation: Ensure models generalize well.

ACCURATE PREDICTION SCORING

- Ensuring reliable predictions (e.g., credit score, churn) that drive business decisions.

- Common Traps:

  - Over-reliance on Accuracy: Use multiple metrics like precision, recall, AUC-ROC, not just accuracy.
  - Class Imbalance: Handle imbalanced datasets with balanced metrics.
  - Overfitting to Historical Data: Ensure the model adapts to evolving patterns.

- How to Improve:

  - Balanced Metrics: Use precision, recall, and AUC-ROC for better assessment.
  - Ensemble Learning: Combine models to improve prediction accuracy.
  - Calibration: Adjust probabilities to improve score accuracy.
  - Model Evaluation: Continuously test and update models to adapt to new data.

THANK YOU!